FIREGUARD AI

Wildfire Response Simulation & Multi-Agency Decision Intelligence

Hack4Guimarães 2035 · Protection & Defence Challenge

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Simulate before you deploy. Coordinate before minutes are lost.

Project description and technical concept for the prototype available at guimaraes-wildfire-watch.lovable.app

IMPORTANT: FIREGUARD AI is a hackathon decision-support prototype. Its results are simulated and are not validated operational forecasts or emergency-response instructions.

1. Executive Summary

FIREGUARD AI is a browser-based wildfire response simulation and decision-support prototype created for the Hack4Guimarães 2035 Protection and Defence challenge. It is designed to help users explore how different response priorities can affect a shared simulated wildfire scenario. Rather than treating an emergency as a static map, the prototype frames it as a changing situation in which weather, terrain, access, resources and community exposure interact.

The application provides a common operating picture, selectable scenario presets, response strategy options, time-step controls, calculated indicators, and a comparison-oriented workflow. The published application describes itself as a common operating picture for a selected simulated scenario and states that displayed figures are computed by the scenario engine. The interface includes fast wind-driven, slope-driven and community-edge fire scenarios, as well as containment, community-protection and balanced-response options.

The goal is not to replace incident commanders or predict a real wildfire. The goal is to make response assumptions visible, support structured discussion, and demonstrate how changing conditions or resource availability can change the relative merits of alternative strategies.

2. The Problem

Wildfire response can involve fire and rescue services, civil protection, police, medical services and municipal authorities. These groups may use different tools, data sources and reporting practices. During a fast-moving incident, teams must build a shared understanding of the fire, available resources, access constraints and communities that may be exposed.

Three practical difficulties motivate the project:

Fragmented situational awareness. Information about weather, roads, settlements and available resources may be spread across different systems or communicated at different times.

Decisions under uncertainty. Decision-makers need to compare options without knowing exactly how conditions will evolve.

Competing priorities. Early containment, community protection, access, and resource availability can conflict. A strategy that performs well against one objective may be weaker against another.

A useful training tool should therefore let teams compare alternatives against the same initial conditions and understand both the results and the assumptions behind them.

3. Proposed Solution

FIREGUARD AI brings scenario inputs, map context, simulated resources and response strategy comparison into one interface. A user selects a scenario, inspects the conditions, chooses or compares response strategies, advances the simulation timeline, and reviews calculated indicators. The workflow is intended to make cause-and-effect easier to discuss during preparedness exercises and demonstrations.

The live prototype is available at https://guimaraes-wildfire-watch.lovable.app/.

4. Application Features

Scenario presets. The published interface provides three starting scenarios: A — Fast wind-driven fire; B — Slope-driven rural fire; and C — Community-edge fire. These give the demo different conditions and planning emphases.

Response strategy selection. The interface offers No response, A — Containment, B — Community, and C — Balanced options. This enables a comparison between different response priorities.

Time-step controls. The dashboard exposes a timeline with selectable 0, 15, 30, 45, 60 and 90 minute points, and a play control. These controls support a time-based walkthrough of the selected simulated scenario.

Operational indicators. The published page displays affected area, burning cells, exposed population and zones, wind speed and direction, humidity, temperature, slope factor, units on scene, and water/medical resources. These values should be understood as scenario-engine outputs rather than live operational observations.

Situation map and access context. The interface includes a situation map and a list of critical access roads such as N101 central axis, Estrada da Penha, N206 east, Variante Sul, N105 Fermentões and a west link near Creixomil. The demo should not imply that road status is live unless an independently verified live data integration exists.

Potential exposure and vulnerable locations. The overview includes potentially exposed zones and illustrative critical locations, including a hospital, school, university and illustrative care-home/school locations. These are for demonstration and planning discussion, not an authoritative safety assessment.

Calculated recommendation and comparison. The published overview shows a calculated recommendation and notes that its ranking changes in a sensitivity test. This is a valuable transparency feature: a recommended strategy should be treated as conditional when plausible changes to assumptions alter the ranking.

Human-led decision support. The prototype is designed to help people compare options. It does not autonomously dispatch responders or issue evacuation orders.

5. Intended Users and Use Cases

Primary users include fire and rescue commanders, civil-protection coordinators, municipal emergency planners, police and traffic coordinators, medical-response planners, and researchers or trainers conducting scenario exercises.

Potential use cases include: (1) training exercises in which participants compare response priorities; (2) preparedness workshops that discuss access and resource constraints; (3) sensitivity analysis showing how changes in wind or available units affect a simulated outcome; (4) multi-agency tabletop exercises using a common incident timeline; and (5) demonstrations of explainable, human-in-the-loop decision-support design.

6. Example Demonstration Flow

A concise demo can be completed in approximately 90 seconds:

Step 1 — Load a scenario. Select the fast wind-driven, slope-driven or community-edge preset and review its starting conditions.

Step 2 — Inspect the common picture. Point out the ignition location, weather indicators, road context, resource availability and potentially exposed zones.

Step 3 — Compare strategies. Run containment, community-protection and balanced response options against the same scenario baseline.

Step 4 — Change one condition. Increase wind or reduce the available resources, then rerun the comparison. Explain which metrics or strategy ranking change.

Step 5 — Discuss uncertainty. Highlight assumptions, the conditional nature of recommendations and the need for professional review.

7. Technical Concept and Data Handling

The prototype is a web application with a scenario-oriented user interface and a simulation engine that computes displayed figures. The public-facing page exposes scenario selection, strategy selection, time controls, indicators and a situation map. The detailed internal implementation should be verified against the current source repository before making claims about the exact framework, model equations, persistence layer, or third-party integrations.

For a responsible demonstration, keep four data categories distinct:

Verified source data: information retrieved from a named, traceable public or institutional source.

User inputs: parameters entered or selected by the person running the scenario.

Synthetic demo data: illustrative roads, resources, ignition points or facilities created for a repeatable demonstration.

Simulated outputs: model-generated footprints, exposure estimates, scores and scenario comparisons.

Potential future data sources include OpenStreetMap for mapped features, public elevation and land-cover datasets, Copernicus emergency-management products, and official Portuguese geographic, weather and civil-protection information where access and terms permit. Any operational use would require data provenance, freshness checks, validation and agreements with relevant authorities.

8. Evaluation and Success Measures

The prototype should be evaluated on whether users can understand the scenario, compare response alternatives and explain why results differ. Useful demo measures include: time needed to load and run a scenario; whether all strategies use identical baseline inputs; whether changes to inputs affect computed results; whether the decision rationale is traceable to visible metrics; whether the output identifies uncertainty; and whether a report can be exported for review.

Do not claim lives saved, validated prediction accuracy, reduced response times or operational effectiveness without a suitable evaluation study and evidence. For a hackathon prototype, the strongest evidence is a repeatable demonstration, clear assumptions and an honest account of limitations.

9. Safety, Limitations and Responsible Use

FIREGUARD AI is not a certified wildfire modelling system, a command-and-control platform, or an emergency dispatch system. A simplified simulation cannot fully represent fuel moisture, changing fire behaviour, spotting, suppression tactics, local microclimates, infrastructure failures, human behaviour or the many uncertainties present in a real incident.

All demo metrics must be labelled simulated. Routes must not be described as verified safe evacuation routes. Illustrative facilities and resource positions must not be represented as confirmed live assets. Recommendations must remain advisory and subject to human judgement. The application should display a persistent training/simulation warning and must not send real alerts or dispatch instructions.

10. Development Roadmap

Stage 1 — Hackathon core: scenario presets, repeatable simulation, interactive timeline, strategy comparison, transparent indicators and a concise demo workflow.

Stage 2 — Data and geospatial quality: documented data imports, terrain and land-cover layers, source metadata, coordinate validation and map-layer controls.

Stage 3 — Decision support: improved resource constraints, sensitivity analysis, scenario replay, audit log and report export.

Stage 4 — Validation: review with domain experts, compare model behaviour with appropriate historical or synthetic benchmark cases, document error bounds and identify safe use limits.

Stage 5 — Institutional pilot readiness: security review, role-based access, privacy and retention review, accessibility, reliability testing, and formal agreements before any operational integration.

11. Conclusion

FIREGUARD AI addresses the Hack4Guimarães 2035 rural-fire challenge by demonstrating how a shared simulated picture and side-by-side response comparison can support more structured emergency planning. Its central value is not an unqualified AI prediction; it is the ability to test alternative decisions, expose trade-offs, and communicate uncertainty in a visual, repeatable way.

Project link: guimaraes-wildfire-watch.lovable.app

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Project prepared for Hack4Guimarães 2035. All operational scenarios and outputs must be treated as simulated unless explicitly verified otherwise.